

IPCC Tier 2 Master Formula

The updated breakdown below incorporates the **exact IPCC Tier 2 mathematical calculations** and the **underlying technical limitations** for each vegetation type.

$$\text{Emissions CO}_2 = A \times MB \times C_f \times EF_{\text{CO}_2}$$

- **A (Area):** 1 hectare (ha) = 10,000m²
- **MB (Biomass Density):** Total available dry organic mass per hectare (tonnes DM/ha)
- **C_f (Combustion Factor):** Fraction of biomass actually consumed during the fire (0.0 - 1.0)
- **EF_{CO2} (Emission Factor):** Conversion factor from burned dry mass to CO₂ (tonnes CO₂ / tonne DM)

Parameters:

- MB = 85 t / ha

- C_f = 0.52

- EF_{CO2} = 1.58 t CO₂ / t DM

Equation for the Coniferous Forest:

$$1 \text{ ha} \times 85 \times 0.52 \times 1.58 = 69.83 \text{ t CO}_2$$

CORINE Land Cover Vegetation Type	CO2 Emitted (Tonnes / Ha)	Real-Life Analogy / Comparison	Direct Source Citation
3.1.2 Coniferous Forest (Maritime Pine / Ponderosa)	50 – 90 t	Burning 1 ha is equal to driving an average gasoline car 360,000 km (~9 laps around the Earth).	Source: ResearchGate / Environmental Management (2011)
			Exact Wording: "An emission factor was not available for mixed forests, which were attributed with the average value of coniferous and broadleaved fuels, i.e. 1,510 g/kg... mean biomass stocks in different compartments: France 118.94 t/ha, Spain 36.9 t/ha"
3.1.1 Broad-leaved Forest (Oak, Chestnut, Beech)	30 – 55 t	Burning 1 ha releases the same annual CO2 as 12 average EU households' total energy use.	Source: IPCC Guidelines for National GHG Inventories (Vol 4, Ch 2 & 4)
			Exact Wording: "Table 2.5: Emission factors for biomass burning... Temperate Forest CO2 EF = 1560 g/kg dry matter burned."

3.1.3 Mixed Forest (Pine & Oak blends)	40 – 70 t	Burning 1 ha is equivalent to Driving 220,000 km in a gasoline car or burning 25,000 kg of coal.	Source: ResearchGate / Environmental Management (2011)
			Exact Wording: "The emission factor from Carvalho et al. (2007) for Mediterranean leaved and mixed forests... 1,510 g/kg."
3.2.1 Natural Grassland (Alpine & Iberian Pastures)	5 – 15 t	Burning 1 ha is like flying 4 round trips from Paris to New York.	Source: Copernicus Earth System Dynamics (2023)
			Exact Wording: "The most important field indicators for the EFs of CO ₂ , CO, and CH ₄ were tree cover density, fuel moisture content, and the grass-to-litter ratio... global savanna/grassland EF_CO ₂ ~1680 g/kg."
3.2.3 Sclerophyllous Vegetation (Maquis, Garrigue, Dehesa)	15 – 35 t	Burning 1 ha equals the annual carbon footprint of 4 European households.	Source: CSIRO / International Journal of Wildland Fire (2013)
			Exact Wording: "Approach 1: 100% of biomass loss... applied by Italy and Spain in the NIR to estimate CO ₂ emissions from forest fires... classified per forest physiognomy."

Key Technical Limitations

1. **Static vs. Dynamic Variables:** Standard Tier 2 models rely on static assumptions for MB and C_f. In reality, C_f fluctuates daily based on the **Fire Weather Index (FWI)** and live fuel moisture.
2. **Gross vs. Net Emissions:** Forest fires release *gross* emissions immediately. Broad-leaved forests and grasslands re-absorb carbon during post-fire recovery over 5–30 years, whereas severe crown fires in coniferous stands can cause long-term soil carbon loss.
3. **Soil Organic Matter (Humus):** These calculations strictly account for **above-ground biomass**. Very intense fires burn into the topsoil/peat layer, releasing up to **20–30% additional unmodeled CO₂**.

	Savanna	Boreal forest	Tropical forest	Temperate forest	Peat	Chaparral	Crop residue	Garbage burning
Carbon dioxide (CO ₂)	1.640×10^3	1.610×10^3	1.625×10^3	1.581×10^3	1.572×10^3	1.649×10^3	1.441×10^3	1.502×10^3
Carbon monoxide (CO)	8.10×10^1	1.00×10^2	1.11×10^2	9.60×10^1	2.25×10^2	6.66×10^1	5.75×10^1	5.20×10^1
Methane (CH ₄)	2.83×10^0	4.78×10	4.68×10^0	4.74×10^0	1.11×10^1	2.57×10^0	2.14×10^0	3.06×10^0
Nitric oxide (NO)	1.76×10^0	9.16×10^{-1}	9.00×10^{-1}	7.85×10^{-1}	3.21×10^{-1}	1.15×10^0	9.62×10^{-1}	8.10×10^{-1}
Nitrogen oxides (NO _x as NO)	3.40×10^0	1.21×10^0	2.55×10^0	1.65×10^0	9.27×10^{-1}	2.42×10^0	2.05×10^0	2.31×10^0
Nitrogen dioxide (NO ₂)	2.60×10^0	9.22×10^{-1}	3.55×10^0	1.40×10^0	5.43×10^{-1}	1.02×10^0	1.96×10^0	2.34×10^0
Nitrous oxide (N ₂ O)	1.40×10^{-1}	2.05×10^{-1}		1.55×10^{-1}		2.50×10^{-1}		
Nitrous acid (HONO)	4.99×10^{-1}	2.55×10^{-1}	1.18×10^0	3.78×10^{-1}	2.22×10^{-1}	5.52×10^{-1}	3.53×10^{-1}	2.51×10^{-1}
Sulfur dioxide (SO ₂)	9.44×10^{-1}	5.64×10^{-1}	4.03×10^{-1}	9.50×10^{-1}	2.06×10^0	5.53×10^{-1}	1.25×10^0	7.05×10^{-1}
Isocyanic acid (CHNO)	1.05×10^0	8.30×10^{-2}		4.05×10^{-1}	5.74×10^{-1}	3.02×10^{-1}	4.69×10^{-1}	1.29×10^{-1}
Ammonia (NH ₃)	6.59×10^{-1}	1.47×10^0	1.33×10^0	1.06×10^0	6.15×10^0	9.09×10^{-1}	9.68×10^{-1}	6.88×10^{-1}
Gaseous non-methane organic compounds (NMOC_g)	3.73×10^1	4.05×10^1	2.53×10^1	4.25×10^1	7.37×10^1	2.17×10^1	3.81×10^1	3.36×10^1
PM _{2.5} ^a	1.6×10^1	1.28×10^1	9.11×10^0	1.79×10^1	2.48×10^1	1.51×10^1	1.27×10^1	9.68×10^0
OA	2.73×10^1 ^b	6.60×10^0		1.71×10^1		1.08×10^1	1.12×10^1	7.36×10^0
OC	6.49×10^0		3.99×10^0	1.04×10^1	1.32×10^1	1.08×10^1	9.47×10^0	5.47×10^0
BC	3.50×10^{-1}	1.30×10^{-1}	3.44×10^{-1}	4.35×10^{-1}	1.60×10^{-2}	6.24×10^{-1}	4.46×10^{-1}	1.98×10^0
EC					4.32×10^{-1}		4.99×10^{-1}	1.92×10^{-1}

In this table (from *Andreae, 2019*, standard biomass burning scientific literature), all values are **Emission Factors (EF)** given in **scientific notation**.

Unit of Measurement

The standard unit for this table is **grams of gas emitted per kilogram of dry biomass burned (g/kg)**.

Converting Scientific Notation (103) to Simple Numbers

The values in the Carbon dioxide (CO₂) row are all multiplied by **103 (1,000)**:

- **1.640×103** =1,640 g of CO₂/kg dry biomass
- **1.581×103** =1,581 g of CO₂/kg dry biomass

What This Means

1.6 kg of CO₂ released for every 1 kg of dry ground vegetation burned

Because CO₂ combines carbon from the fuel with oxygen (O₂) from the air, the weight of the emitted gas is **higher** than the weight of the dry plant material itself!

CO₂ Values by Ecosystem (Simple Values)

Land / Vegetation Type	Value in Image	Simple Number (g CO ₂ /kg)	Tonnes of CO ₂ per Tonne of Biomass Burned
Savanna	1.640×103	1,640 g/kg	1.64 tonnes
Boreal forest	1.610×103	1,610 g/kg	1.61 tonnes
Tropical forest	1.625×103	1,625 g/kg	1.63 tonnes
Temperate forest (France/Spain)	1.581×103	1,581 g/kg	1.58 tonnes
Peat	1.572×103	1,572 g/kg	1.57 tonnes
Chaparral (Maquis/Shrub)	1.649×103	1,649 g/kg	1.65 tonnes
Crop residue	1.441×103	1,441 g/kg	1.44 tonnes

Garbage burning	1.502×10^3	1,502 g/kg	1.50 tonnes
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Key Takeaway

Notice how **consistent** CO₂ emissions are across biomes: roughly **1.58 to 1.65 tonnes of CO₂** for every 1 tonne of dry organic matter consumed.

This confirms why forest density (MB) and burned area (ha) are the main drivers of total emissions: while the emission rate per unit mass burned is nearly constant (~1.6 t CO₂/t biomass), a pine forest holds significantly more biomass per hectare than a grassland.